

License Expiry Management and Notification System i

1. AGENT APPLICATION FLOW (MOBILE)

1.1 Authentication

- User opens application
- Enters credentials (mobile + OTP or password)
- System validates via backend authentication service
- Session stored locally
- Redirect to Dashboard

1.2 Attendance Management

Punch In

- Capture timestamp and GPS
- Save record

Punch Out

- Capture timestamp and GPS
- Close attendance record

1.3 Shop Creation

- Capture shop image
- Auto capture GPS location
- Optional shop details input
- Save:
 - Image URL
 - Latitude/Longitude

- Agent ID
- Timestamp

1.4 Shop Visit and License Renewal

- Select shop

If renewal pending or nearing expiry:

- Show alert notification

Renewal Process:

- Enter payment amount
- Capture selfie
- Add remarks
- Submit

System Actions:

- Save renewal record
- Update license duration
- Update expiry date
- Notify Admin

1.5 License Expiry Awareness (Agent Side)

Expiry Conditions

- Each shop has:
 - License Start Date
 - License Duration (e.g., 1 year)
 - Expiry Date

System Behavior

- If current date is close to expiry (e.g., within 7 days):
 - Show warning in dashboard
- If expired:
 - Mark shop as “Expired”
 - Highlight in red in UI

1.6 Reminder Logic (Enhanced)

- When agent is near shop (GPS < 100m):
 - Check:
 - Renewal pending OR
 - License near expiry OR
 - License expired

Trigger:

- Show reminder:
 - “License expiring soon”
 - “License expired – renew immediately”

1.7 Dashboard

Displays:

- Attendance status
- Total shops
- Pending renewals
- Expiring licenses
- Expired licenses

1.8 Offline Handling

- Store all actions locally
- Sync when online
- Maintain consistency using timestamps

2. ADMIN APPLICATION FLOW (WEB)

2.1 Authentication

- Secure admin login
- Redirect to dashboard

2.2 Dashboard

Displays:

- Total agents
- Active agents
- Total shops
- Pending renewals
- Expiring licenses
- Expired licenses

2.3 Agent Management

- Create/update agents
- Assign shops
- Monitor attendance

2.4 Shop Management

- View shops with:
 - Image
 - Location
 - Agent
 - License details

License Fields Displayed:

- Start Date
- Duration
- Expiry Date
- Status:
 - Active
 - Expiring Soon
 - Expired

2.5 Renewal Management

- View renewal requests

Approval Flow:

- Validate:
 - Payment
 - Selfie proof
- Approve / Reject

On Approval:

- Update:
 - New license start date
 - New duration
 - New expiry date
 - Status = Active

2.6 Attendance Monitoring

- Daily attendance logs
- Geo validation
- Exception tracking

2.7 Notification System (Enhanced)

Trigger Events

- New shop added
- Renewal submitted
- Renewal approved/rejected
- License nearing expiry
- License expired

Recipients

- Agent (mobile push)
- Admin (dashboard alerts)

3. LICENSE EXPIRY MANAGEMENT LOGIC

3.1 Data Fields (Add to Shops Table)

license_start_date
license_duration_days
license_expiry_date
license_status (active / expiring / expired)

3.2 Expiry Calculation Logic

license_expiry_date = license_start_date + license_duration_days

3.3 Status Rules

Condition	Status
Today < expiry - 7 days	Active
Today within 7 days of expiry	Expiring Soon
Today > expiry date	Expired

3.4 Automated Jobs (Cron / Scheduled)

Run daily:

FOR each shop:

IF today > expiry_date:
 status = expired

ELSE IF today >= expiry_date - 7 days:
 status = expiring

3.5 Notification Logic

Expiring Soon

- Trigger: 7 days before expiry
- Notify:

- Agent assigned to shop
- Admin

Expired

- Trigger: On expiry date
- Notify:
 - Agent
 - Admin

4. END-TO-END FLOW (UPDATED)

1. Agent logs in
2. Agent punches in
3. Agent visits shop
4. System checks:
 - Renewal pending
 - License expiring/expired
5. Agent performs renewal
6. Renewal submitted
7. Admin reviews
8. Admin approves
9. License duration updated
10. Expiry recalculated
11. Notifications sent
12. Cycle continues

5. DATABASE UPDATE (FINAL STRUCTURE)

Shops Table (Updated)

shops

- id
- name
- image_url
- latitude
- longitude
- created_by

- license_start_date
- license_duration_days
- license_expiry_date
- license_status
- created_at

6. DEVELOPMENT PLAN (UPDATED)

Phase 1–3 (Same as earlier)

Phase 4 (Enhanced)

- Implement expiry calculation logic
- Add cron job for daily status update
- Add notification triggers for:
 - Expiring
 - Expired

7. UI/UX REQUIREMENTS (IMPORTANT)

Agent App

- Highlight shops:
 - Yellow → Expiring soon
 - Red → Expired

Admin Panel

- Filters:
 - Active
 - Expiring
 - Expired

8. SYSTEM OPTIMIZATION IMPACT

With expiry automation:

- Reduces manual tracking effort
- Ensures timely renewals
- Improves compliance visibility
- Enables proactive agent actions

9. FINAL NOTES

- License lifecycle must be system-driven, not manual
- Notifications should be proactive, not reactive
- Expiry logic should run automatically via scheduled jobs
- All renewal approvals must update license lifecycle immediately